EART11300

Igneous, sedimentary and metamorphic rocks in hand specimen

Video 2: Attributes of **sedimentary** rocks

Sedimentary rocks

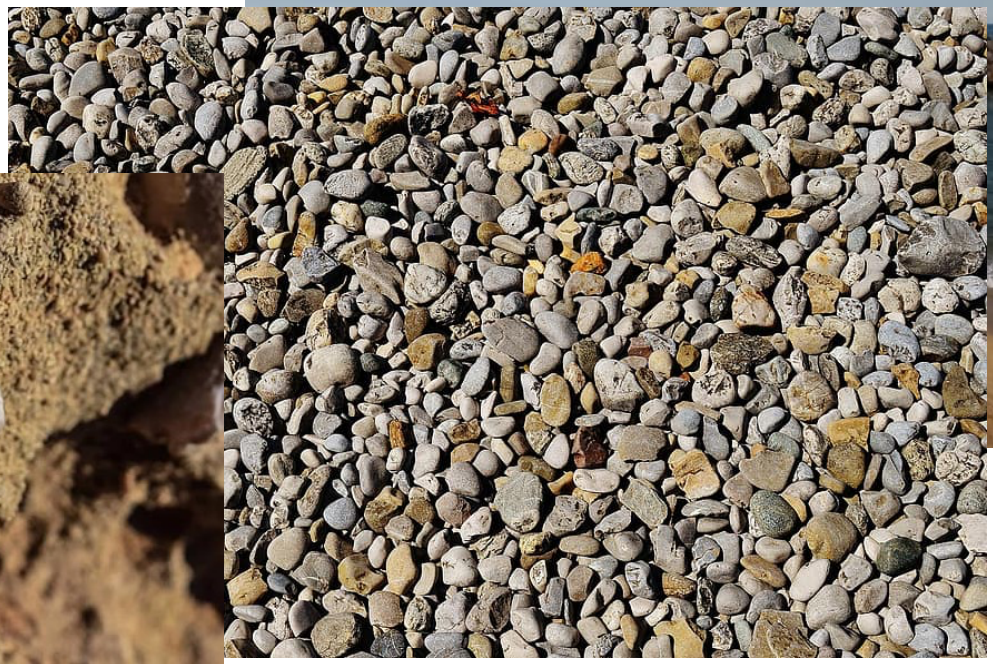
When you are looking at a hand specimen or thin section:

- Commonly consist of particles (clasts) of pre-existing rocks, which may include polycrystalline rock fragments.
- Clasts may be rounded, indicating abrasion during sediment transport.
- May contain fossils.
- May contain organic matter.
- May be quite porous.
- Commonly layered (bedded), often with dramatic compositional contrasts. (May only see this when looking at a large scale, e.g. on maps or in the field.)

Sedimentary rocks

- Commonly consist of particles (clasts) of pre-existing rocks, which may include polycrystalline rock fragments.
- Before becoming a solid rock, these clasts could have been, e.g.:
  - Sand on the beach
  - Pebbles in a river
  - Shells





Sedimentary rocks

- Commonly consist of particles (clasts) of pre-existing rocks, which may include polycrystalline rock fragments.
- The clasts may be fragments of igneous rock, metamorphic rock or a pre-existing sedimentary rock







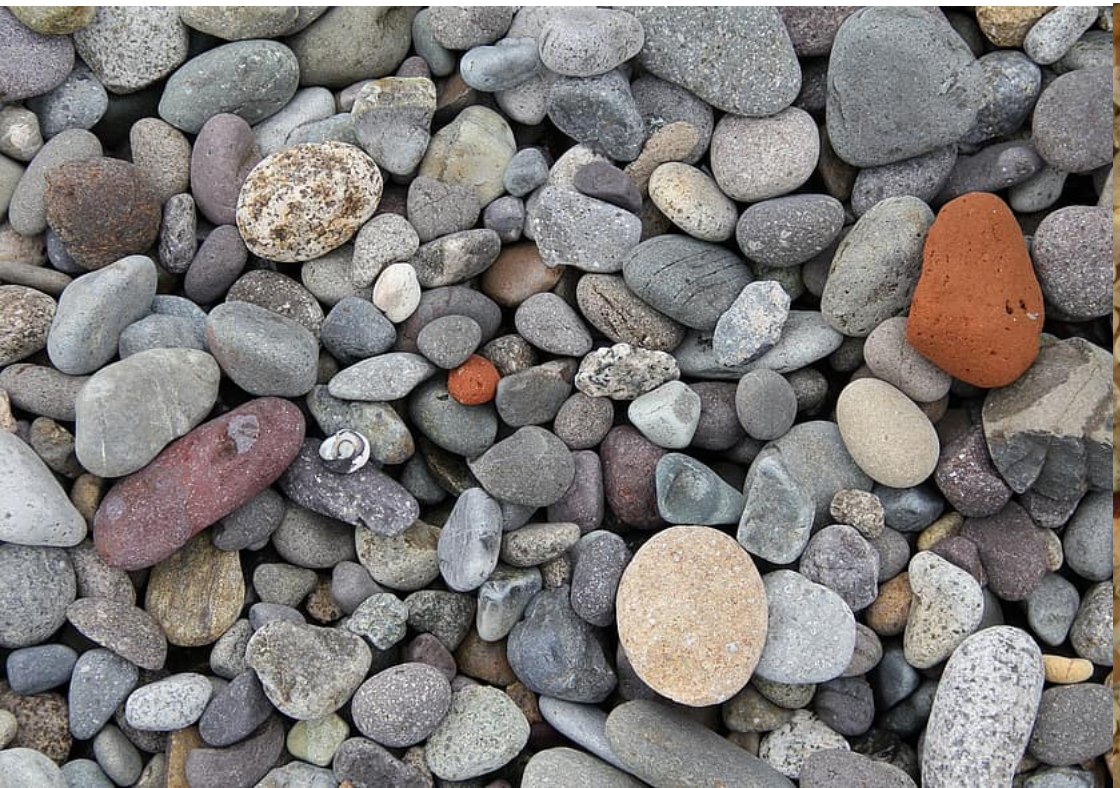
Sedimentary rocks

- Commonly consist of particles (clasts) of pre-existing rocks, which may include polycrystalline rock fragments
- or they may be single crystals derived from one of these rock types
- or they may be fossils (or bits of organic matter)





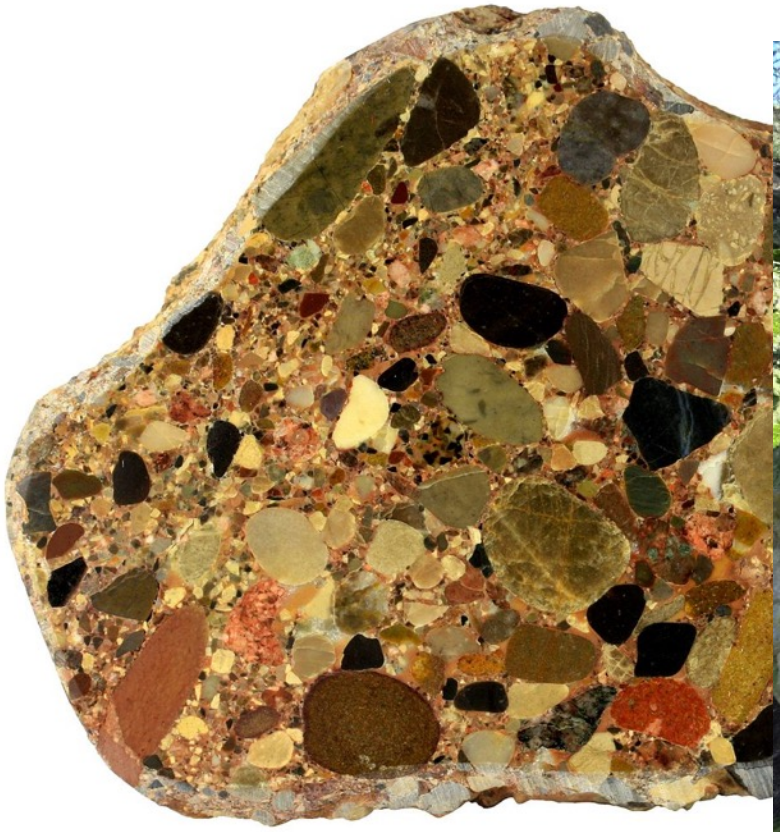
- Clasts may be rounded, indicating abrasion during sediment transport.
- They are also usually weathered – look at the dull, whitish surfaces of many of these pebbles – you are not likely to see sparkly crystals.
- Note that quartz is resistant to weathering, but grains may become coated with e.g. iron oxide (rust).





Sedimentary rocks

- May be quite porous:
- In a sedimentary rock made of clasts, the clasts are held together by a cement. There are often gaps in this cement = porosity.





Sedimentary rocks

- Commonly layered (bedded), often with dramatic compositional contrasts. (May only see this when looking at a large scale, e.g. on maps or in the field.)





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can be classified
the distance travelled

grain size.

grain size.

ergy available for

